

ECON 1550 International Finance

Output and the Exchange Rate in the Short Run

Behavioral equations in the goods market

Demand for domestic goods

Equilibrium in the goods market

A short-run model of the goods market

Exogenous variables

Variable	Description
E	Nominal exchange rate
I	Investment
G	Government spending
T	Taxes
P	Price level
P^*	Foreign price level
Y^*	Foreign income

Endogenous variables

Variable	Description	Equation	Type of equation
Y	Income, production	$Y = D$	Equilibrium condition
Y_D	Disposable income	$Y_D \equiv Y - T$	Identity
EX	Exports	$EX = EX(q, Y^*)_{(+)(+)}$	Behavioral
IM	Imports	$IM = IM(q, Y_D)_{(-)(+)}$	Behavioral
CA	Current account	$CA \equiv EX - IM = CA(q, Y_D, Y^*)_{(+)(-)(+)}$	Identity
D	Demand for domestic goods	$D \equiv C + I + G + CA$	Identity
A	Domestic demand	$A \equiv C + I + G$	Identity
C	Consumption	$C = C(Y_D)_{(+)}$	Behavioral
q	Real exchange rate	$q \equiv \frac{EP^*}{P}$	Identity

DD Curve

$$Y = D$$

$$Y = C + I + G + CA$$

$$Y = \underset{(+)}{C}(Y_D) + I + G + \underset{(+)}{CA}(\underset{(-)}{q}, \underset{(+)}{Y_D}, Y^*)$$

$$Y = \underset{(+)}{C}(Y_D) + I + G + \underset{(+)}{CA}(\underset{(-)}{EP^*/P}, \underset{(+)}{Y_D}, Y^*)$$

Short-run FX and money market model

Exogenous variables

Variable	Description
R^*	Foreign interest rate
E^e	Expected exchange rate
Y	Real income
M^s	Money supply
P	Price level

Endogenous variables

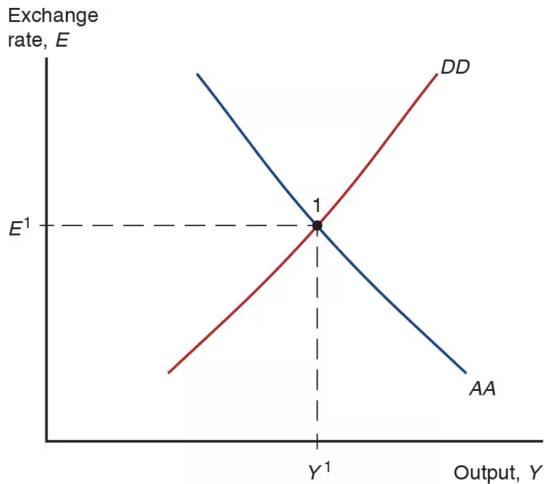
Variable	Description	Equation	Type of equation
E	Exchange rate	$R = R^* + \frac{E^e}{E} - 1$	Equilibrium condition
R	Domestic interest rate	$M^d/P = L(R, Y)$	Behavioral equation
M^d	Money demand	$M^d = M^s$	Equilibrium condition

AA Curve

$$\text{(UIP):} \quad R = R^* + \frac{E^e}{E} - 1$$

$$\text{(MS) = (MD):} \quad \frac{M^s}{P} = L(R, Y)$$

Short-run Equilibrium in $AA-DD$ model



Example: DD Schedule

$$C_{(+)}(Y_D) = 1 + 0.75Y_D$$

How to find the DD schedule

$$EX_{(+)}(q, Y^*)_{(+)} = 0.15 + 0.5q + 0.1Y^*$$

$$IM_{(-)}(q, Y_D)_{(+)} = 0.1 - 0.2q + 0.12Y_D$$

$$Y_D = Y - T$$

$$q = \frac{EP^*}{P}$$

Example: AA Schedule

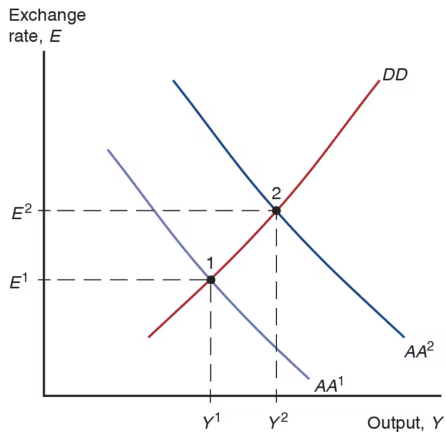
$$R = R^* + \frac{E^e}{E} - 1$$

How to find the AA schedule

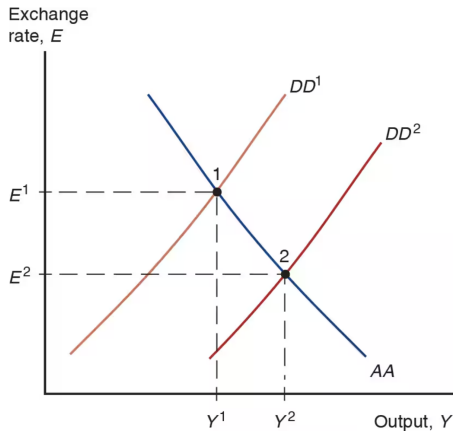
$$\frac{M^d}{P} = \frac{M^s}{P}$$

$$\begin{aligned}\frac{M^d}{P} &= L_{(-)}(R, Y_{(+)}) \\ &= 1.1 - R + 0.05Y\end{aligned}$$

Temporary Change in Monetary Policy



Temporary Change in Fiscal Policy



Example: DD Schedule

$$C(Y_D) = 1 + 0.75Y_D$$

(+)

$$EX(q, Y^*) = 0.15 + 0.5q + 0.1Y^*$$

(+)

$$IM(q, Y_D) = 0.1 - 0.2q + 0.12Y_D$$

(-)

$$Y_D = Y - T$$

$$q = \frac{EP^*}{P}$$

- Plug in given C , EX , IM , Y_D , q into the equilibrium condition for the goods market $Y = C + I + G + EX - IM$ to get:

$$Y = 1.05 + 0.63(Y - T) + 0.7 \frac{EP^*}{P} + I + G + 0.1Y^*$$

- Solve for E to get the DD curve:

$$E = \frac{1}{0.7} \frac{P}{P^*} (-1.05 + 0.37Y + 0.63T - I - G - 0.1Y^*)$$

Example: AA Schedule

$$R = R^* + \frac{E^e}{E} - 1$$

$$\frac{M^d}{P} = \frac{M^s}{P}$$

$$\frac{M^d}{P} = L\left(\underset{(-)}{R}, \underset{(+)}{Y}\right) = 1.1 - R + 0.05Y$$

- Plug in UIP into money market equilibrium condition:

$$\frac{M^s}{P} = 1.1 - \left(R^* + \frac{E^e}{E} - 1\right) + 0.05Y$$

- Solve for E to get the AA curve:

$$E = \frac{E^e}{2.1 - \frac{M^s}{P} - R^* + 0.05Y}$$

Full AA-DD Model

DD Schedule: $Y = C(Y - T) + I + G + CA(EP^*/P, Y - T, Y^*)$

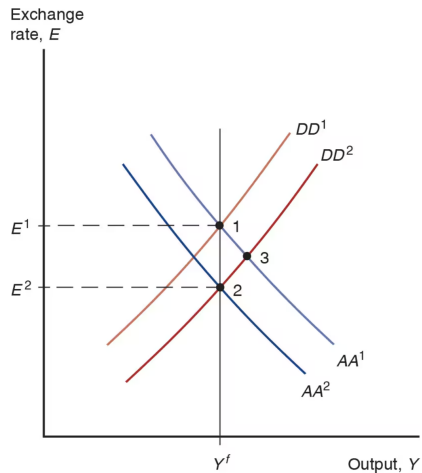
AA Schedule: $\frac{M^s}{P} = L\left(R^* + \frac{E^e}{E} - 1, Y\right)$

Phillips Curve: $\pi = \pi^e + \alpha(Y - Y^f)$

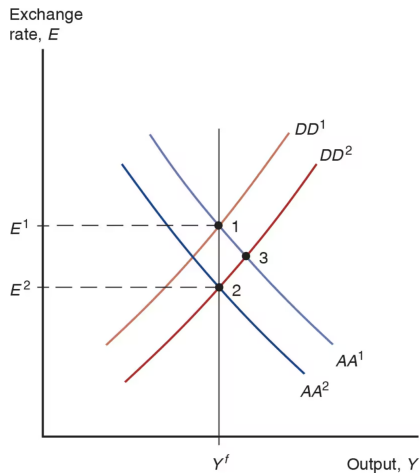
Definition of inflation: $\pi_t = \frac{P_t}{P_{t-1}} - 1$

Definition of expected inflation: $\pi_t^e = \frac{P_{t+1}^e}{P_t} - 1$

Permanent Shifts in Fiscal Policy

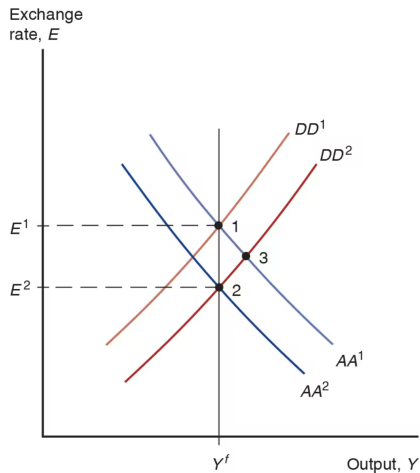


Permanent Shifts in Fiscal Policy



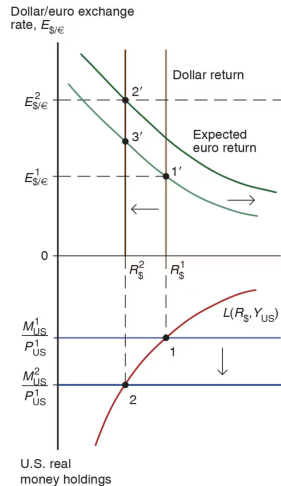
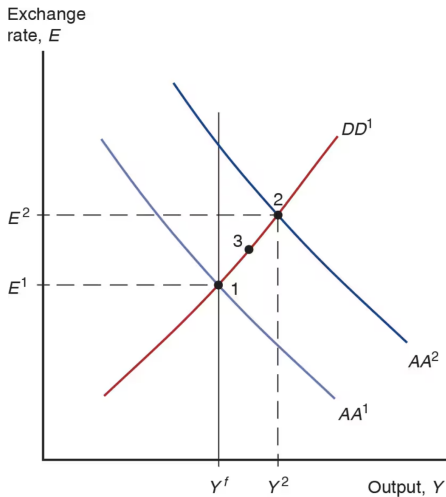
- Start in a medium run equilibrium at point 1 with $R_0 = R^*$ and $E_0^e = E^1$
- Increase in G shifts DD to the right
- At point 3, exchange rate is lower than E_1
- Since the shift in DD is permanent, appreciation at point 3 is also expected to be permanent
- E^e must also go down
- Lower E^e shifts AA down
- But how much?

Permanent Shifts in Fiscal Policy



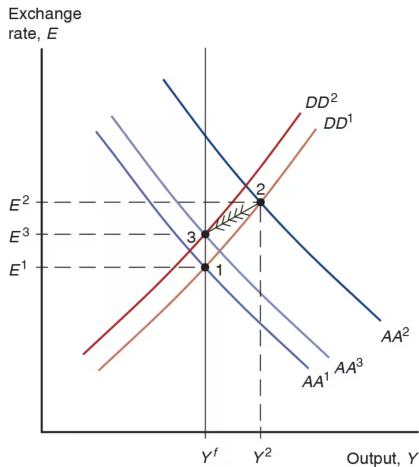
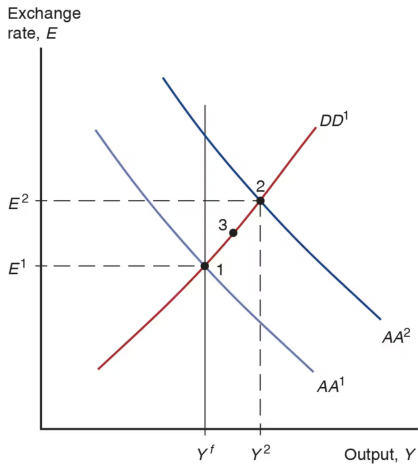
- There are no more shifts expected to occur, E^e remains constant
- Real money supply is constant
- Long run real money demand must stay unchanged
- Then P never changes
- $MS=MD$ implies R does not change
- UIP implies E does not change either
- If Y must be equal to its original level Y^f , AA^1 must shift to AA^2

Permanent Shifts in Monetary Policy

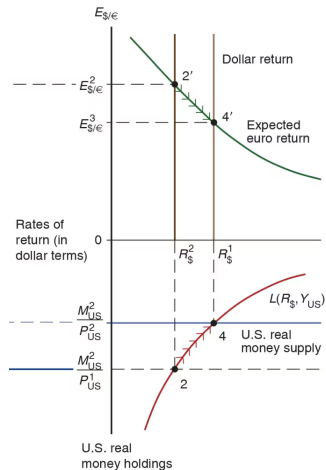
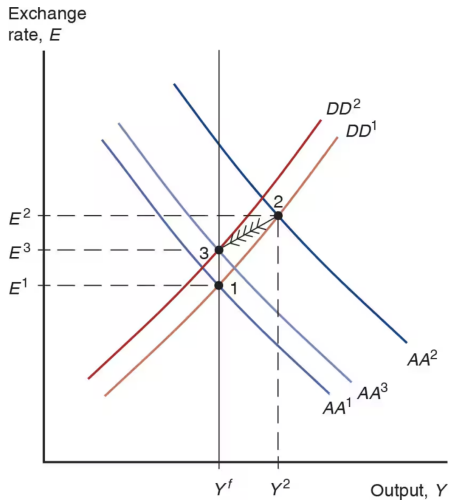


(a) Short-run effects

Permanent Shifts in Monetary Policy



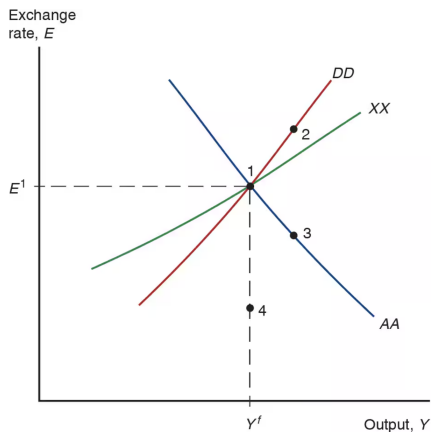
Permanent Shifts in Monetary Policy



(b) Adjustment to long-run equilibrium

DD Schedule with Tariffs

The XX Curve



- Pick a fixed value $\bar{X}$.
- The XX schedule gives the combinations of E and Y that have

$$CA \left(\underset{(+)}{EP^*/P}, \underset{(-)}{Y - T}, \underset{(+)}{Y^*} \right) = \bar{X}.$$

- XX is upward sloping.
- XX is flatter than DD .
- $CA > \bar{X}$ above XX , and $CA < \bar{X}$ below XX .

Why is XX flatter than DD?

- XX schedule: $CA(\underset{(+)}{EP^*/P}, \underset{(-)}{Y - T}, \underset{(+)}{Y^*}) = \bar{X}$
- DD schedule: $Y = c_0 + c_1(Y - T) + I + G + CA(EP^*/P, Y - T, Y^*),$
with $c_1 < 1$
- If Y and E increase so that $CA = \bar{X}$ (along XX), then
 $Y + \Delta Y > c_0 + c_1(Y + \Delta Y - T) + I + G + \bar{X}.$
- To restore goods-market equilibrium, CA must rise, so E must rise.